

Communication systems

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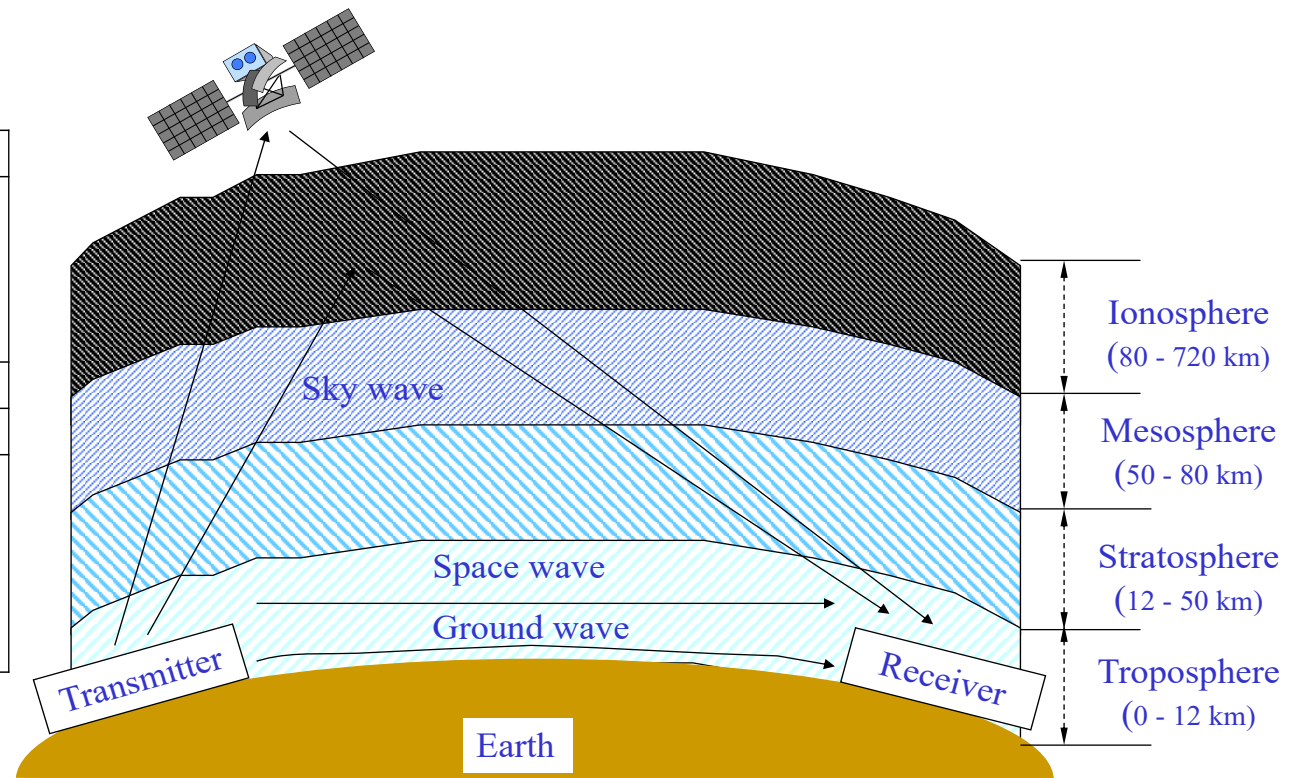
ELECTRONICS AND COMMUNICATIONS SYSTEMS

COMPUTER ENGINEERING

Wireless propagation channel

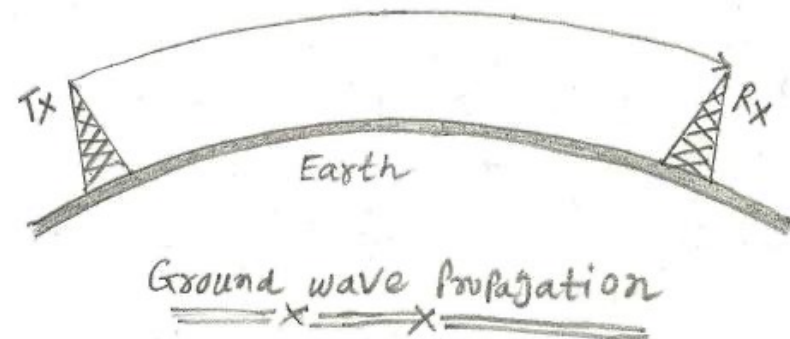
Signal propagation in the air

Classification Band	Initials	Frequency Range	Characteristics
Extremely low	ELF	< 300 Hz	Ground wave
Infra low	ILF	300 Hz - 3 kHz	
Very low	VLF	3 kHz - 30 kHz	
Low	LF	30 kHz - 300 kHz	
Medium	MF	300 kHz - 3 MHz	Ground/Sky wave
High	HF	3 MHz - 30 MHz	Sky wave
Very high	VHF	30 MHz - 300 MHz	Space wave
Ultra high	UHF	300 MHz - 3 GHz	
Super high	SHF	3 GHz - 30 GHz	



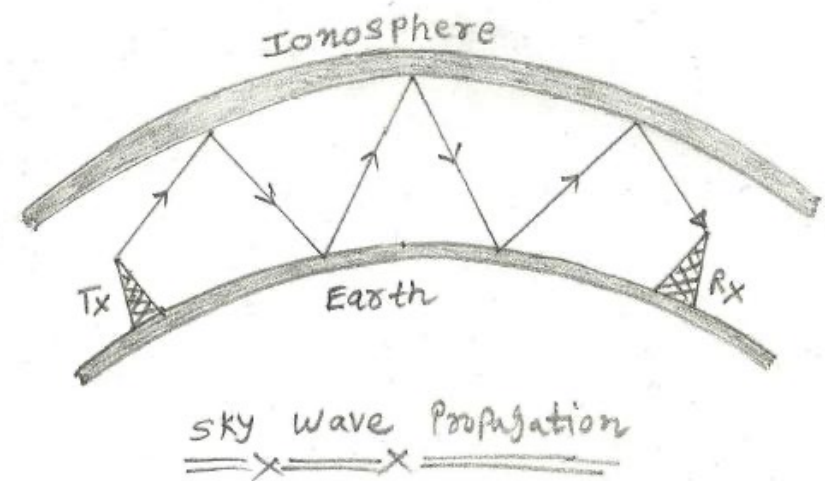
Ground wave propagation

- The wave propagates following earth's curvature reaching receivers beyond the horizon (in certain cases hundreds of km).
- Valid for frequencies below 2 MHz, LF-MF bands



Sky wave propagation

- For a certain ranges of frequency (around 10 MHz) the ionosphere acts as a mirror and reflects the signals back to the earth.
- Bouncing between earth and the ionosphere the signal can propagate up to a few thousands of km.
- Valid mainly for HF.



Space wave propagation

- For frequencies larger than 30 MHz, the main form of propagation is line-of-sight.
- The received signal is composed by the direct component plus the paths reflected by nearby obstacles.
- The larger the frequency the larger is the propagation attenuation.

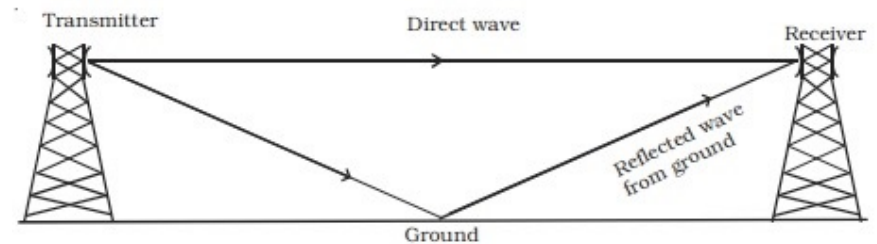
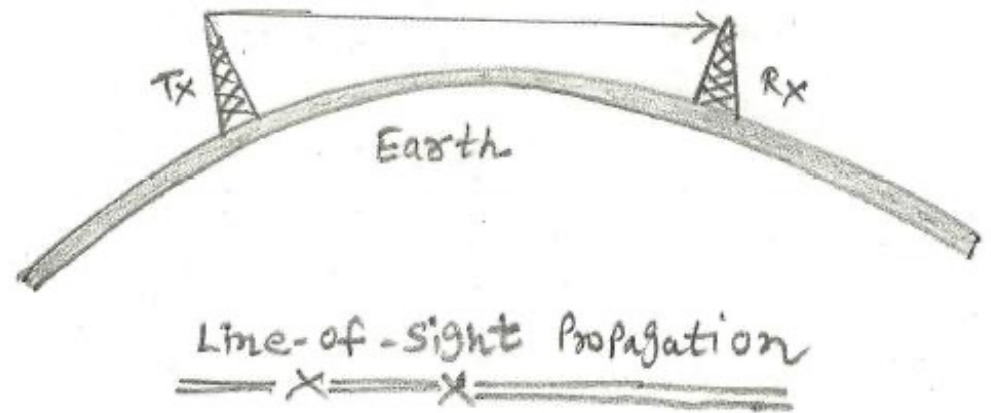
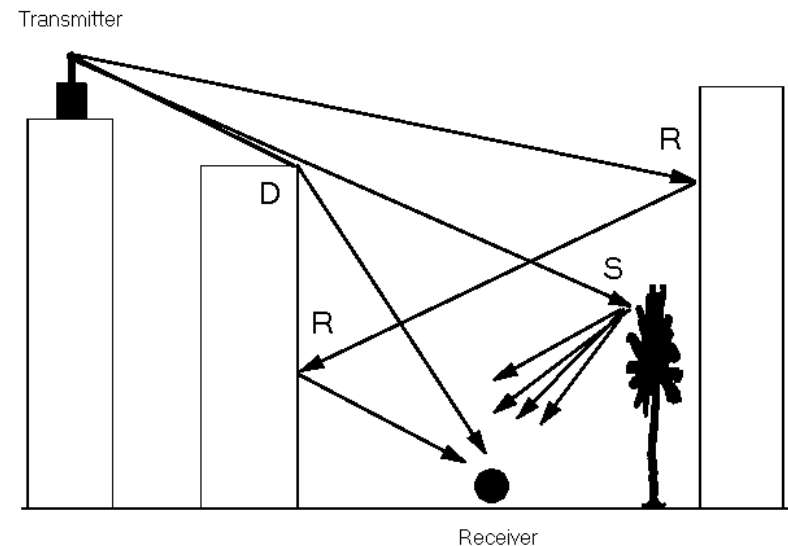


Fig Space wave propagation

The wireless propagation channel (space wave)

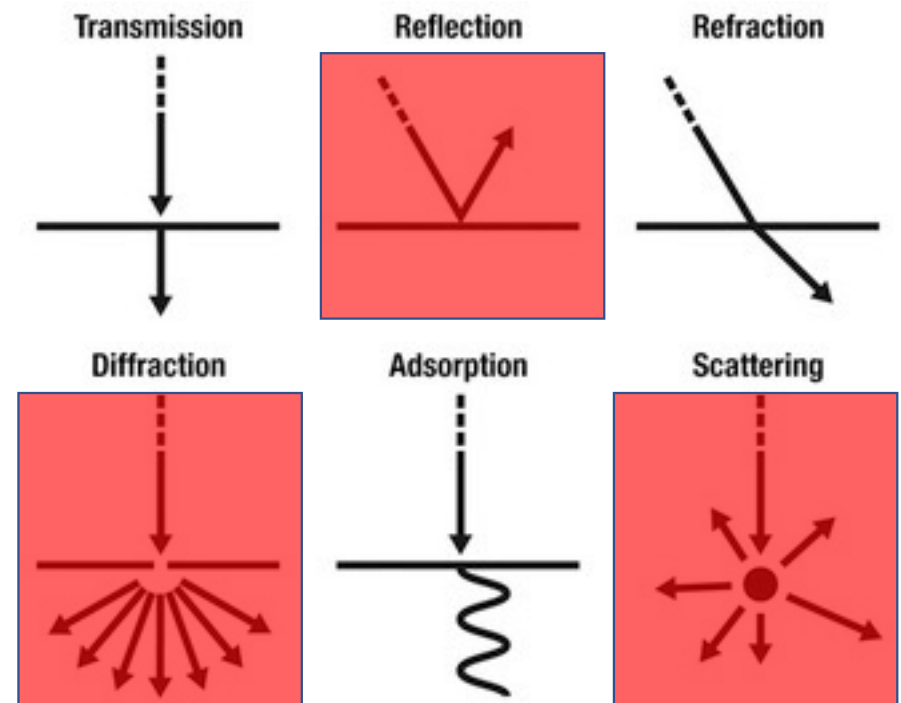
- Because, mobile services are mostly in the bandwidth 30 MHz-30 GHz, *spacewave* is the most important wave propagation mechanism we need to consider.
- The main physical phenomena are: reflection, diffraction, scattering.
- The effect of the combination of these propagation phenomena can be summarized into *large-scale* and *small-scale* fading.



Reflection (R), diffraction (D) and scattering (S).

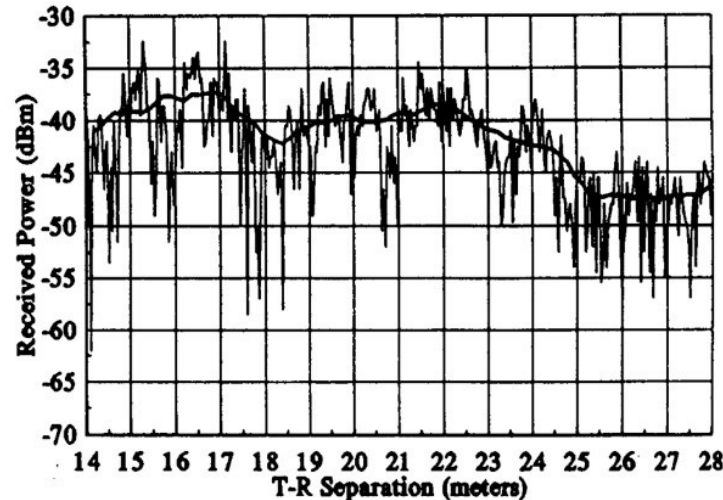
Propagation phenomena

- Three major propagation mechanisms:
 - **Reflection**
Signal impinges on very large (w.r.t. to signal wavelength) objects. When a wave meets a boundary, it can be either reflected or transmitted.
 - **Diffraction**
Signal is obstructed by objects that have sharp irregularities. Diffraction depends on the size of the object relative to the wavelength of the wave.
 - **Scattering**
Propagation medium populated by small (wrt to signal wavelength) objects or rough surfaces (e.g. foliage, street signs).



Large-scale fading

- Large-scale fading: propagation models that characterize average signal strengths over Tx-Rx separation distance.
- Accounts for averaged received power, changes over distances ≈ 1 m.
- Large-scale fading can be modelled as the combination of *path-loss* and *shadowing*.



Physics of wireless signal propagation

- Electromagnetic waves travel at speed of light spreading out in all directions

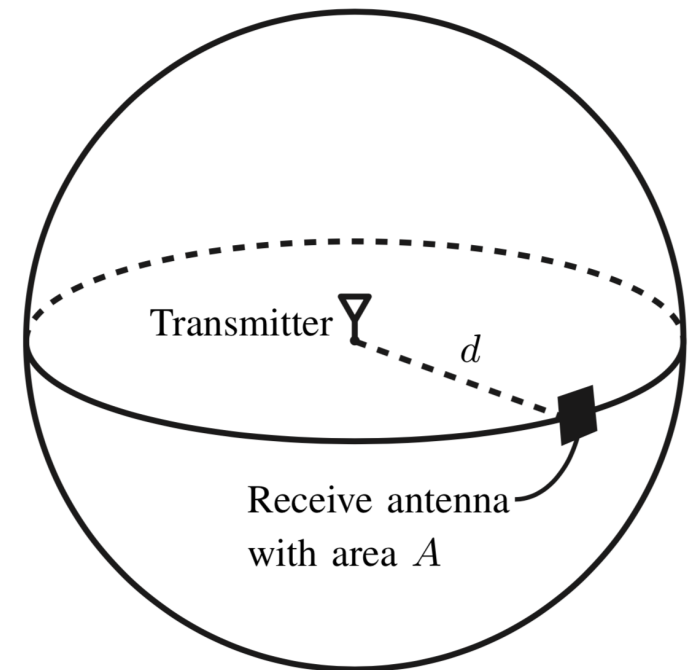
$$\text{Receive power} = \text{Transmit power} \cdot \frac{A}{4\pi d^2}$$

Example: $A = \frac{1}{\pi} \left(\frac{\lambda}{2}\right)^2$ $\lambda = 0.1 \text{ m}$ (3 GHz)

0.006% received at 1 m (−42 dB)

0.00006% received at 10 m (−62 dB)

Tiny fraction of the transmit power is received!



Large-scale fading: path-loss

- Path-loss models simplify Maxwell's equations.
- Models vary in complexity and accuracy but, in general, mean power falloff w.r.t. the tx-rx distance d is proportional to d^2 in free space and to d^n in other environments.
- Considering only path-loss, the average received signal power is

$$P_{Rx} \propto P_{Tx} \Gamma(f_0, d_0) \left(\frac{d_0}{d} \right)^n \quad d > d_0$$

- *Near field term* $\Gamma(f_0, d_0) \approx \left(\frac{\lambda}{4\pi d_0} \right)^2$

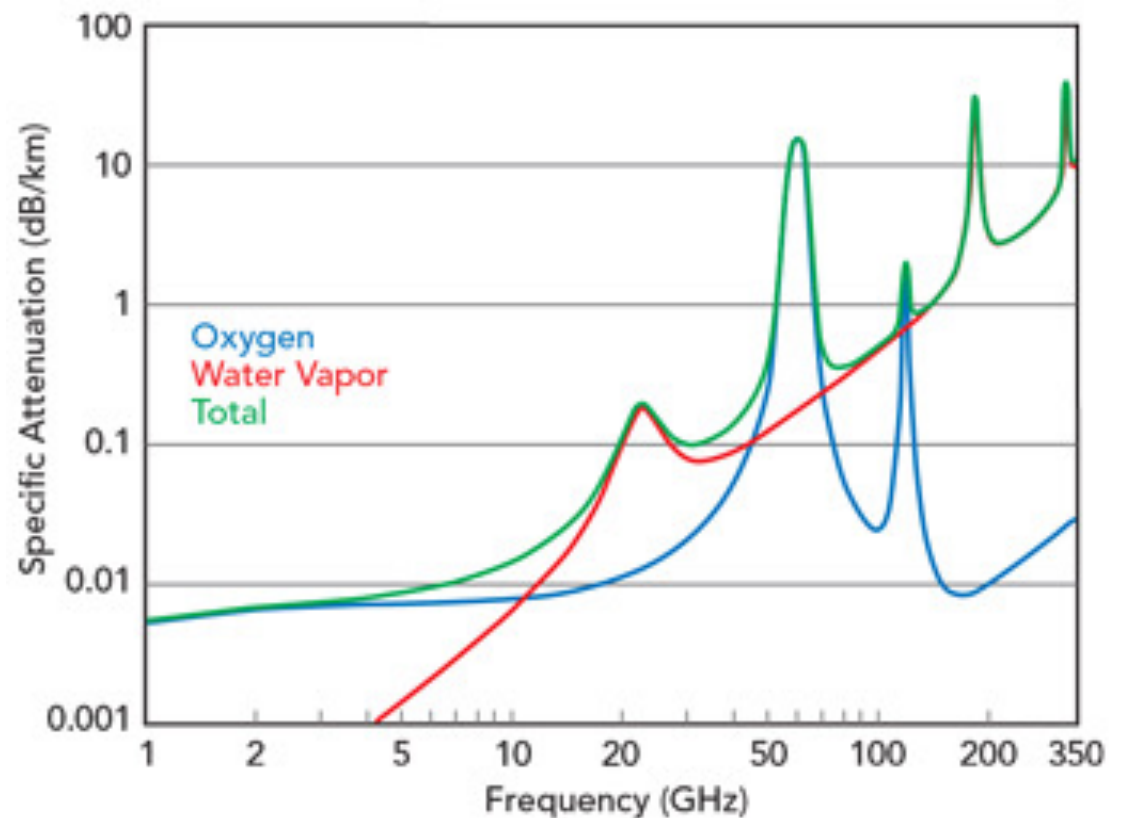
- Path-loss A_{PL} is

$$A_{PL} = \frac{P_{Tx}}{P_{Rx}} = \Gamma(f_0, d_0) \left(\frac{d_0}{d} \right)^n$$

Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 – 3.5
Urban area cellular (obstructed)	3 – 5
In-building line-of-sight	1.6 – 1.8
Obstructed in-building	4 – 6
Obstructed in-factories	2 – 3

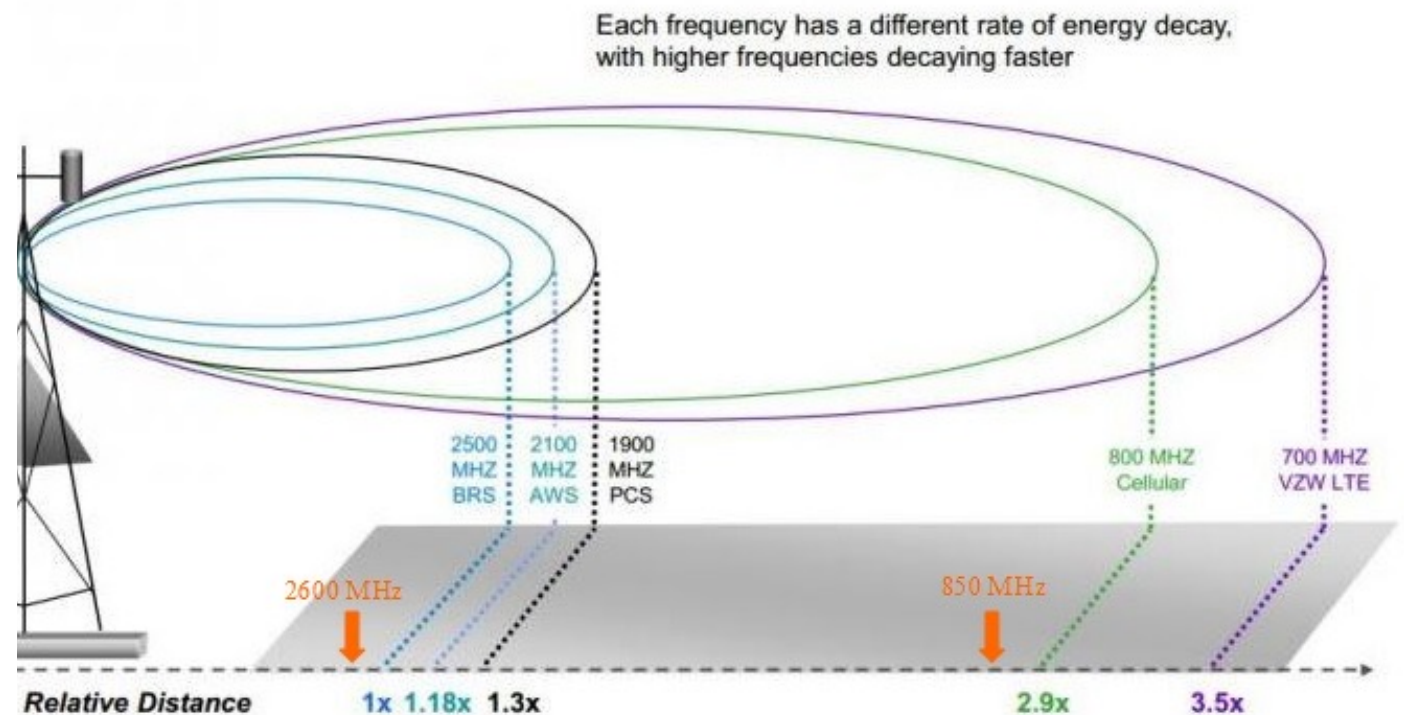
Large-scale fading: attenuation due to frequency

- At frequencies $f_c < 6$ GHz ($\lambda > 50$ mm) the channel attenuation depends on the carrier frequency with a square law.
- At larger frequencies the channel attenuation depends on other physical phenomena such as oxygen and water vapor absorption.
- Mmwave channel is very much attenuated!



Path-loss and cell size

- At cell borders path-loss attenuation may exceed 100 dB.
- The larger the carrier frequency the larger is the attenuation and the smaller is the cell radius.
- Large cells, designed for *coverage*, use low carrier frequencies, small or very small cells, designed to boost *capacity*, use mmwave frequencies.



Large-scale fading: shadowing

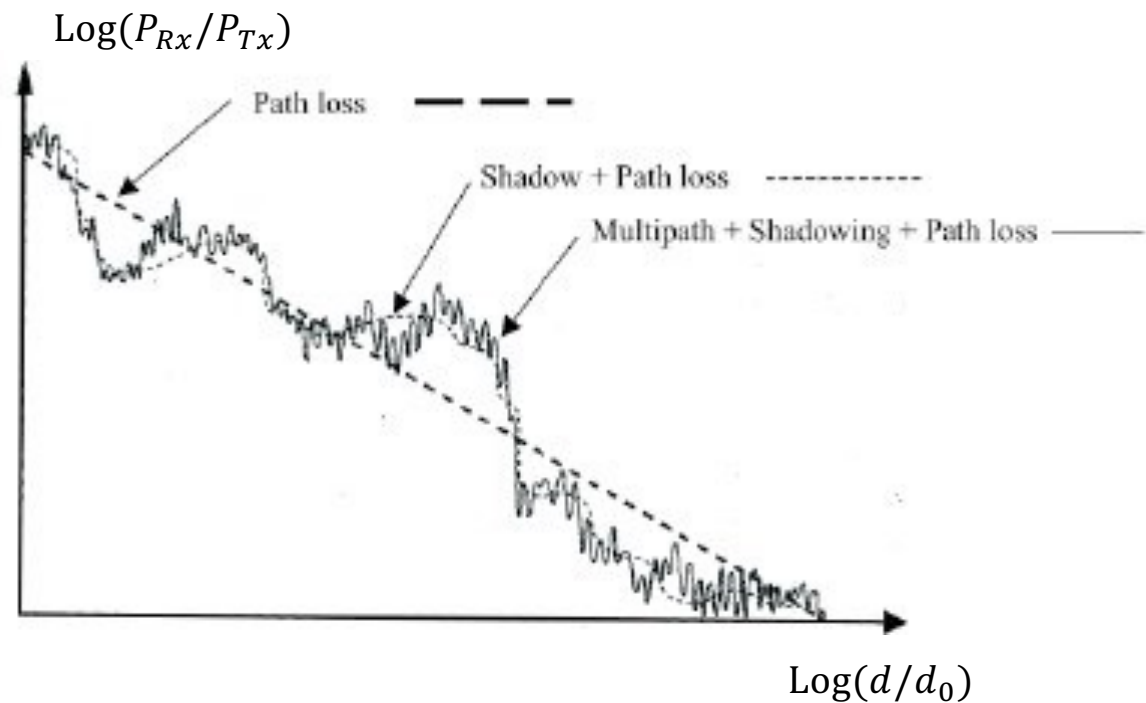
- Two points with the same distance d from the transmitter have theoretically the same path-loss, nevertheless their average attenuation may still greatly differ.
- *Shadowing* accounts for the random variations of the average channel attenuation.
- Shadowing A_S is a random variable *log-normally distributed* with parameters $\mu = 0$ and σ_S , expressed in dB.
- The pdf in dB of A_S is

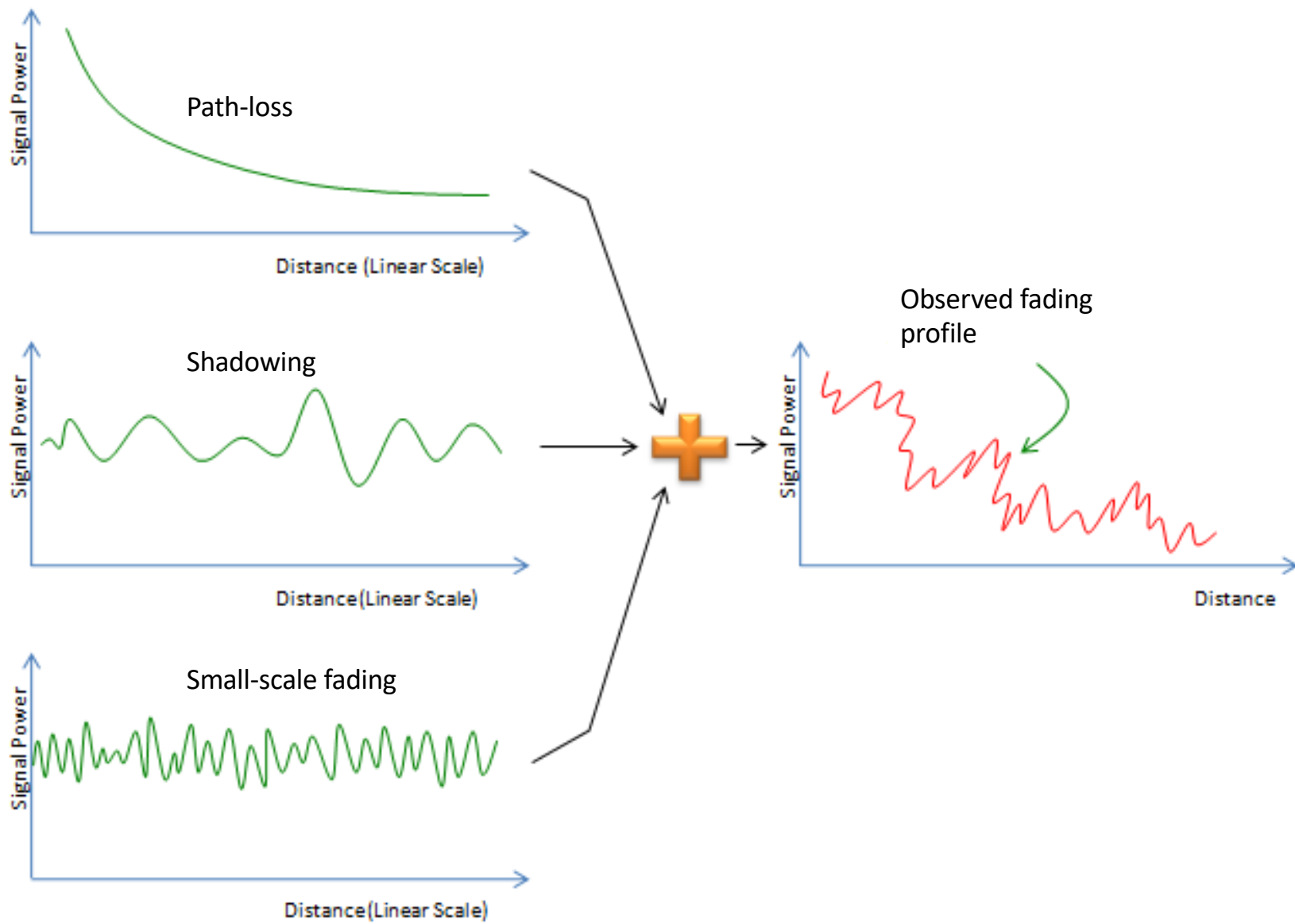
$$p(A_S) = \frac{1}{\sqrt{2\pi}\sigma_S} e^{-\frac{A_S^2}{2\sigma_S^2}}$$

where σ_S is the standard deviation in decibels (typical values 0-9 dB)

Large-scale fading

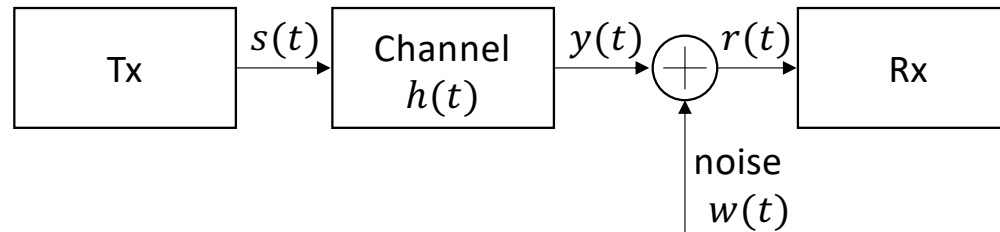
- The received power in dB is computed as
$$P_{Rx}[dBm] = P_{Tx}[dBm] + \underbrace{A_{PL}[dB] + A_S[dB]}_{A_{LS}} + A_{SS}[dB]$$
- A_{PL} is deterministic and depends on the distance d .
- A_S is a log-normally distributed random variable.
- A_{SS} is the attenuation due to small scale fading, which fluctuates rapidly with the distance.





Propagation channel: small scale fading

- The propagation channel can be modeled as a LTI filter.

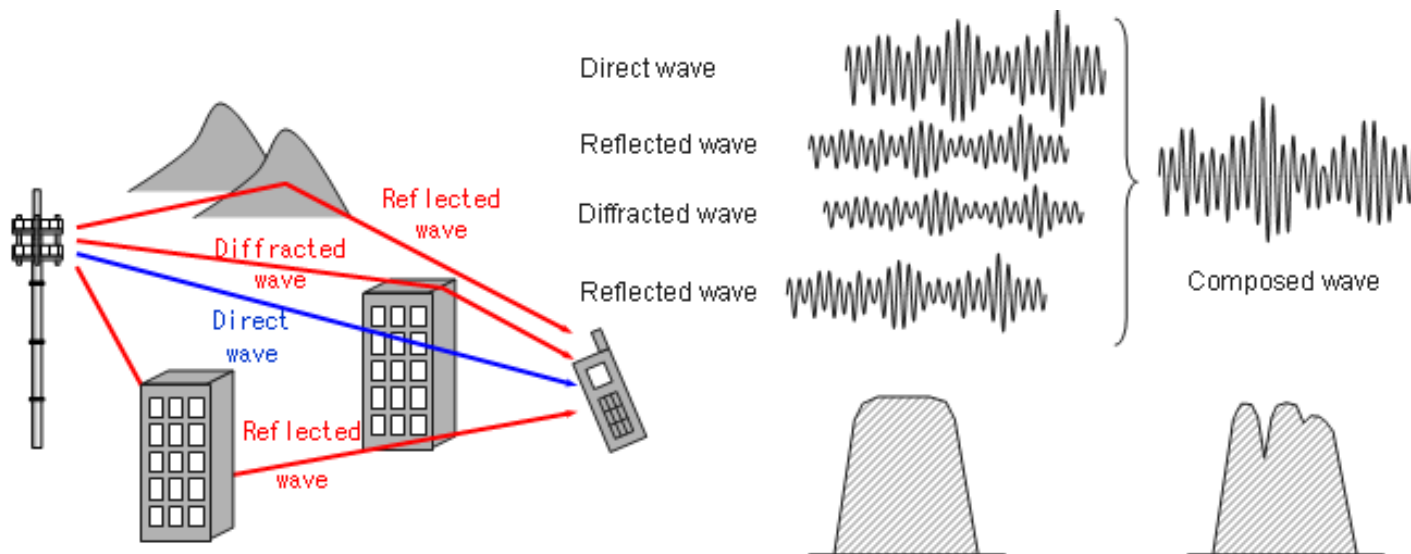


- The filter impulse response $h(t)$ depends on the *small-scale fading* characteristics.

$$y(t) = s(t) \otimes h(t)$$

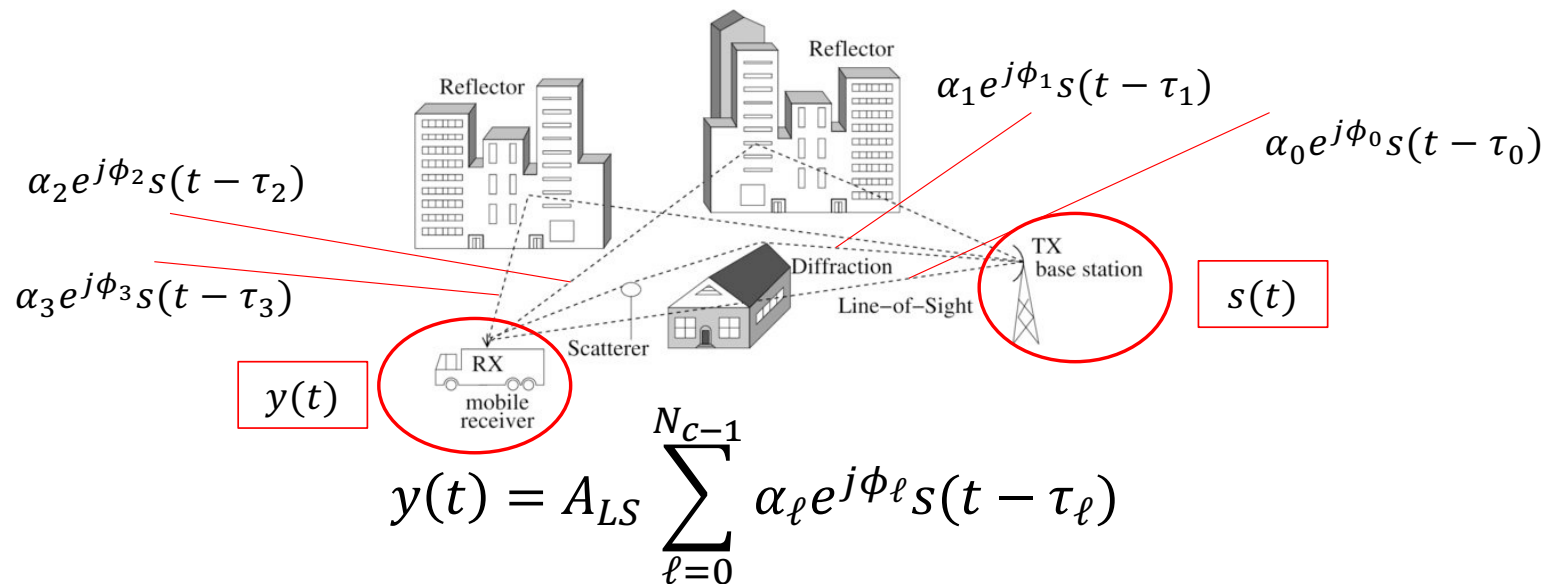
Small-scale fading

- Small-scale fading: accounts for the random variations of the instantaneous received power over distances of the order of a wavelength.
- Because of the various propagation phenomena, a large number of waves, each carrying a replica of the transmitted signal, arrives at the receiver.



Small-scale fading

- Neglecting the noise, the complex envelope of the signal at the receiver is the sum of delayed replicas of the transmitted signal $s(t)$, each with its own delay phase and attenuation, i.e.



Small-scale fading: Rayleigh distribution

- Considering that a τ -delayed replica can be seen as the convolution of $s(t)$ with $\delta(t - \tau)$, the impulse response of the channel can be represented as

$$h(t) = A_{LS} \sum_{\ell=0}^{N_e-1} \alpha_{\ell} e^{j\phi_{\ell}} \delta(t - \tau_{\ell})$$

Path-loss and shadowing

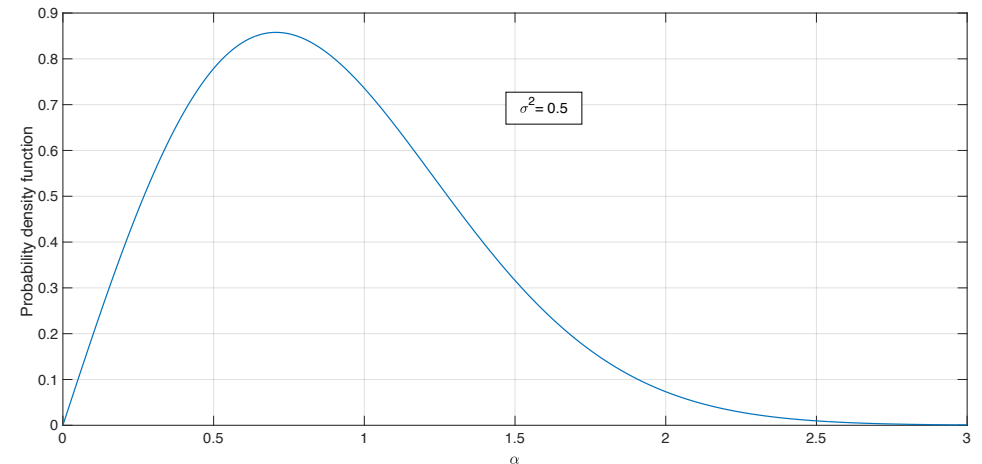
Multipath

- The path gains α_{ℓ} are modeled as random variables. In most cases, they follow a Rayleigh distribution.
- The path phases ϕ_{ℓ} are modelled as uniformly distributed variables in the interval $[0, 2\pi]$.

Channel gain characterization

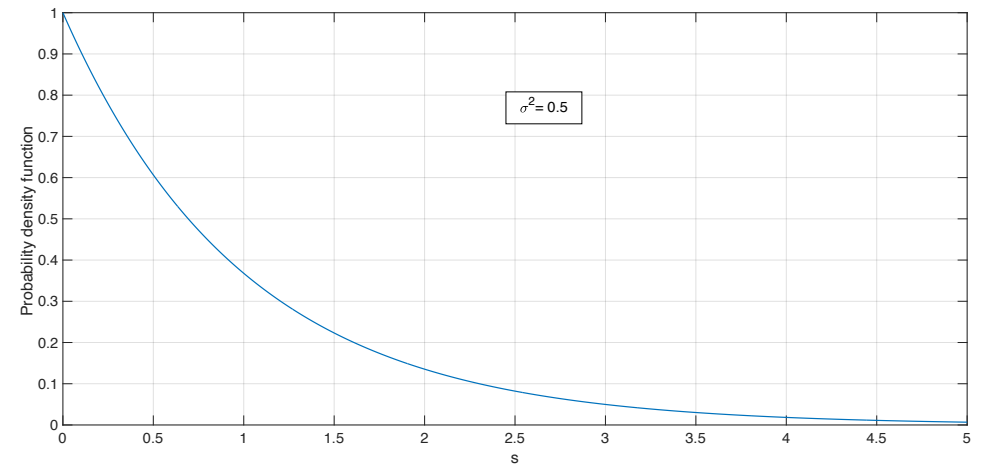
- The distribution for channel amplitude α is *Rayleigh*

$$p(\alpha) = f_A(\alpha) = \begin{cases} \frac{\alpha}{\sigma^2} e^{-\frac{\alpha^2}{2\sigma^2}} & \alpha \geq 0 \\ 0 & \alpha < 0 \end{cases}$$



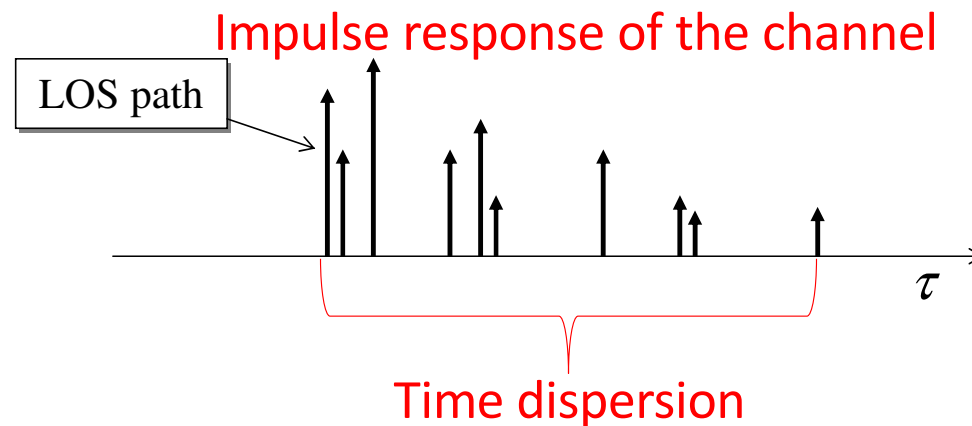
- The distribution for channel power $s = \alpha^2$ is *exponential*

$$p(s) = f_S(s) = \begin{cases} \frac{1}{2\sigma^2} e^{-\frac{s}{2\sigma^2}} & s \geq 0 \\ 0 & s < 0 \end{cases}$$



Small-scale fading: multipath channel

- The various signal's replicas may interfere with each other and generate inter-symbol (ISI) interference.



- The channel's time dispersion is measured by the *delay spread* σ_τ .

Channel's delay spread

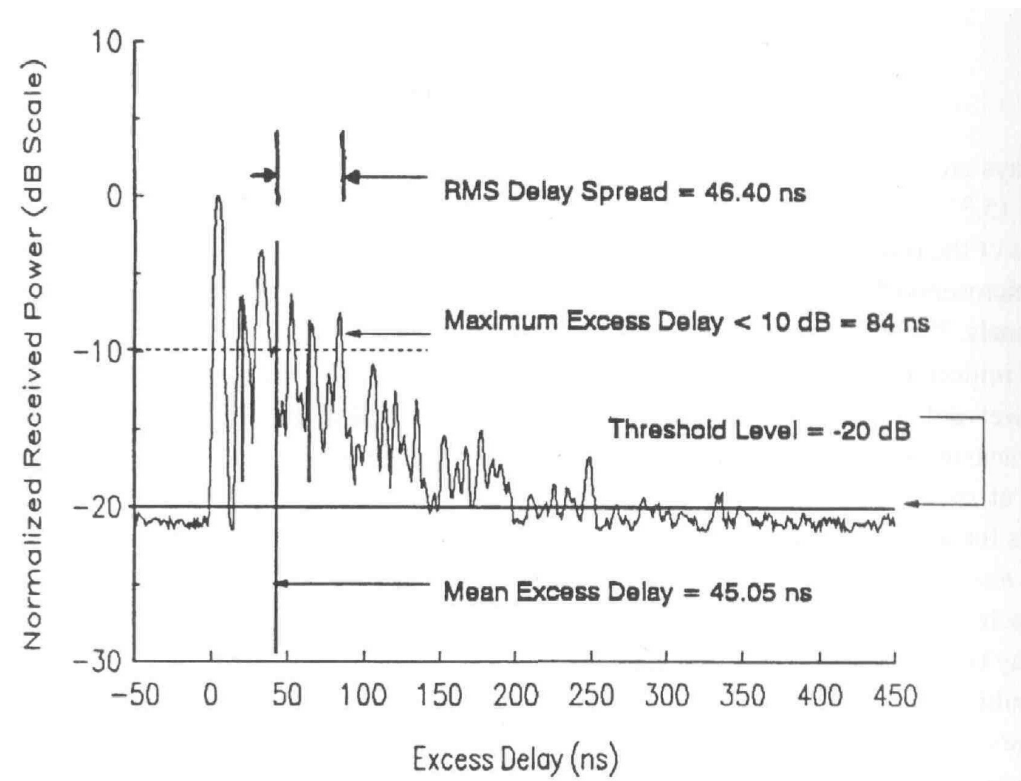
- To compute the delay statistics, one should integrate over the density function of the delays.
- Too difficult, a practical method consist in weighting the delay of each path by the coefficients $0 < \alpha_\ell^2 \setminus \sum_{\ell=0}^{L-1} \alpha_\ell^2 < 1$, which are equivalent to empirical mass probabilities.
- *Mean excess delay*

$$\bar{\tau} = E\{\tau\} = \int_0^{+\infty} \tau \, pdf(\tau) d\tau \approx \sum_{\ell=0}^{N_c-1} \frac{\alpha_\ell^2}{\sum_{\ell=0}^{L-1} \alpha_\ell^2} \tau_\ell ;$$
$$\overline{\tau^2} = \sum_{\ell=0}^{N_c-1} \frac{\alpha_\ell^2}{\sum_{\ell=0}^{L-1} \alpha_\ell^2} \tau_\ell^2$$

- *RMS delay spread*

$$\sigma_\tau^2 = E\{(\tau - \bar{\tau})^2\}$$
$$\sigma_\tau \approx \sqrt{\overline{\tau^2} - \bar{\tau}^2}$$

RMS delay spread



Multipath channel in frequency

- Neglecting the noise and the path-loss ($A_{LS} = 1$), the output of the channel is

$$y(t) = s(t) \otimes h(t)$$

and the spectrum of the received signal

$$Y(f) = S(f)H(f),$$

where $H(f)$ is the transform of $h(t) = \sum_{\ell=0}^{N_c-1} \alpha_{\ell} e^{j\phi_{\ell}} \delta(t - \tau_{\ell})$, i.e.

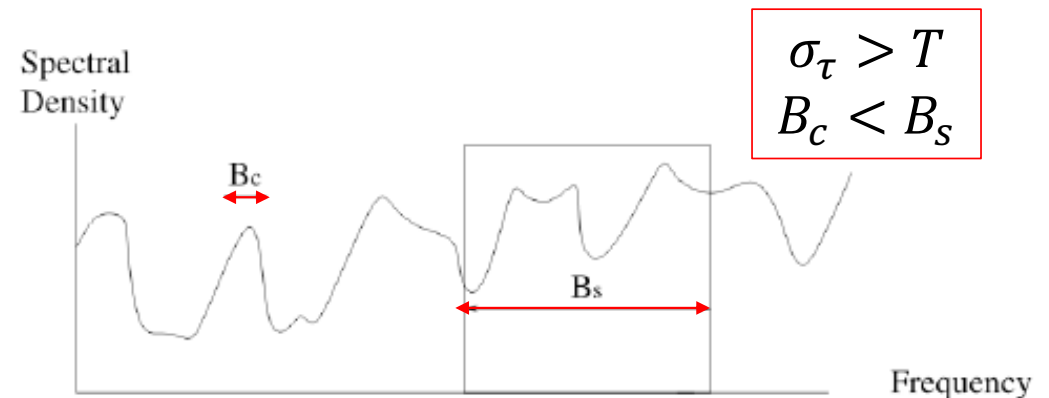
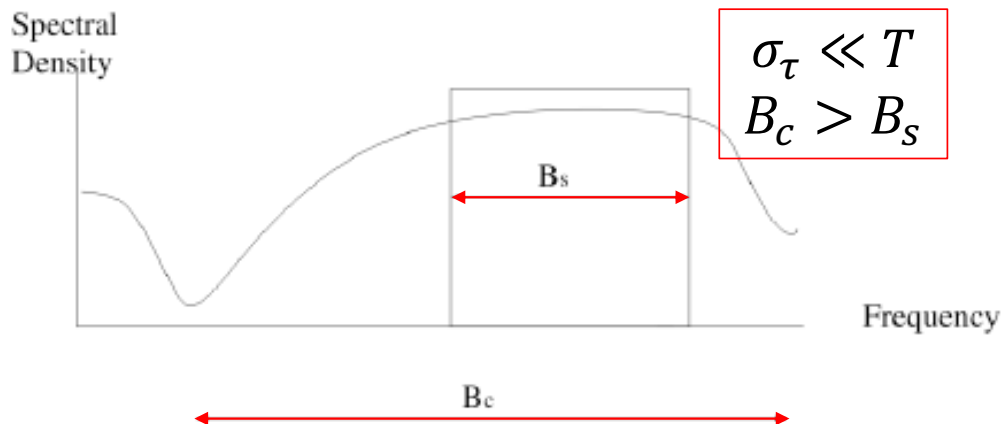
$$H(f) = \sum_{\ell=0}^{N_c-1} \alpha_{\ell} e^{j\phi_{\ell}} e^{-j2\pi f \tau_{\ell}}$$

Coherence bandwidth

- The channel *coherence bandwidth* B_c is the frequency interval over which the channel's frequency response $H(f)$ (and its power spectral density $S_h(f)$) is approximately constant.

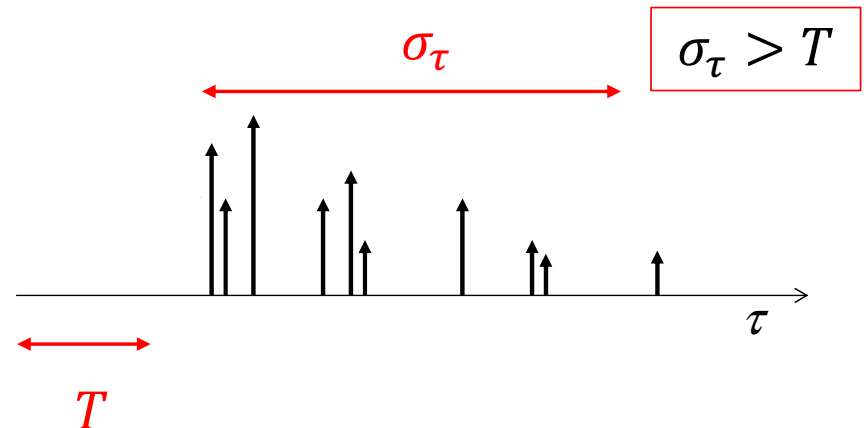
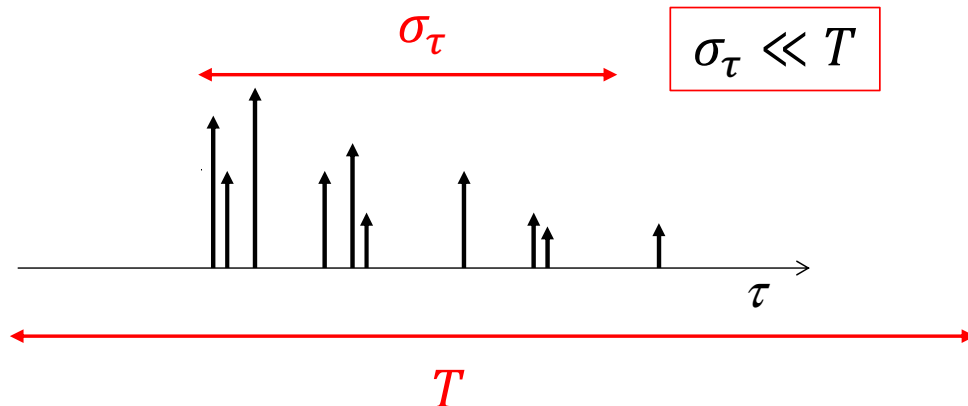
$$B_c \approx \frac{1}{5\sigma_\tau}$$

- When $\sigma_\tau \ll T$, it is $B_c > B_s$, the channel is frequency flat or just flat.
- When $\sigma_\tau > T$, it is $B_c < B_s$, the channel is frequency selective (multipath).



Time dispersion – delay spread

- If it is $\sigma_\tau \ll T$, the multipath channel can be approximated with only one resolvable path and the channel is *flat* fading.
- If $\sigma_\tau > T$, there are more than one resolvable path and the channel is *multipath* and *frequency selective*. The replicas of the received signal interfere with each other.



Small-scale fading: multipath channel

- Neglecting the noise and the path-loss ($A_{LS} = 1$), the receive filter's output is

$$x(t) = \sum_i c_i g(t - iT)$$

where $g(t) = g_T(t) \otimes h(t) \otimes g_R(t) = g_N(t) \otimes h(t)$ so that it is

$$g(t) = \sum_{\ell=0}^{N_c-1} \alpha_\ell e^{j\phi_\ell} g_N(t - \tau_\ell).$$

- In the multipath channel, the decision variable $x(m)$ is

$$x(m) = x(t) \Big|_{t=mT} = \sum_k c_{m-k} g(kT) = c_m g(0) + \sum_{k, k \neq 0} c_{m-k} g(kT)$$

where it is

$$g(kT) = g(t) \Big|_{t=kT} = \sum_{\ell=0}^{N_c-1} \alpha_\ell e^{j\phi_\ell} g_N(kT - \tau_\ell)$$

Typical values of RMS delay spread

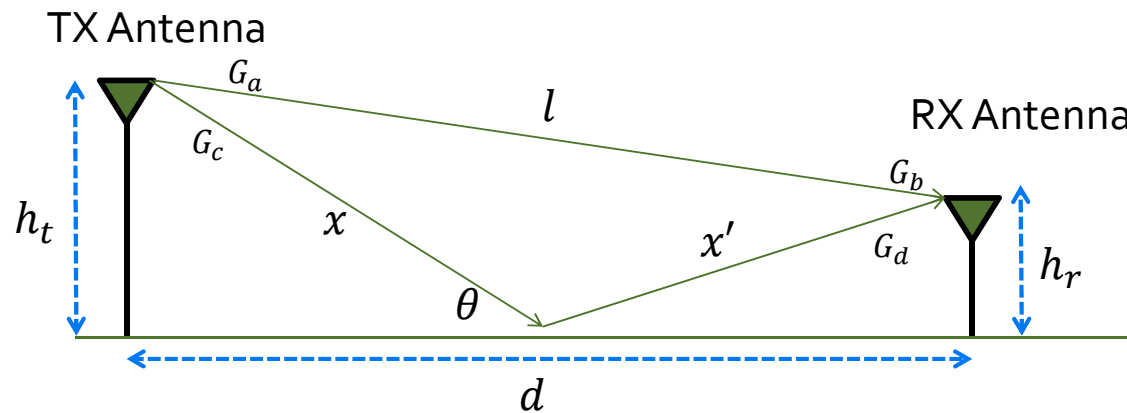
- Delay spread depends on signal frequency and type of environment.
- Typical values of delay spread are 0.2μs (rural area), 0.5μs (suburban area), 3-8μs (urban area), <2 μs (urban microcell) and 50-300ns (indoor picocell)
- A delay spread of 5 μs corresponds to the coherence bandwidth B_c

$$B_c \approx \frac{1}{5} \frac{1}{5 \cdot 10^{-6}} = 40\text{kHz!}$$

Environment	RMS delay spread (σ_τ)	Coherence bandwidth (B_c)	Notes
Urban	1300 ns	150 kHz	NYC
LTE ETU	Up to 5 μs	40 kHz min	Averaged typical case
Suburban	2000 ns	100 kHz	Averaged extreme case
Indoor	10-100 ns	2-20 MHz	Office building

Example: the two-ray channel model

- Channel parameters



$$h(t) = \alpha_0 \delta(t - \tau_0) + \alpha_1 \delta(t - \tau_1)$$

$$H(f) = \alpha_0 e^{-j2\pi f \tau_0} + \alpha_1 e^{-j2\pi f \tau_1}$$

- $\alpha_0 = 1$
- $\alpha_1 = 0.9$
- $\tau_0 = 0$
- $\tau_1 = \tau$
- $p_0 = \frac{1}{1+0.9^2} \approx 0.55$
- $p_1 = \frac{0.81}{1.81} \approx 0.45$

Delayed since $x+x'$ is longer. $\tau = (x + x' - l)/c$

$$r_{2\text{-ray}}(t) = \text{Re} \left\{ \frac{\lambda}{4\pi} \left[\frac{\sqrt{G_a G_b} \tilde{g}(t) \exp\left(-\frac{j2\pi l}{\lambda}\right)}{l} + \frac{R \sqrt{G_c G_d} \tilde{g}(t - \tau) \exp\left(-\frac{j2\pi(x + x')}{\lambda}\right)}{x + x'} \right] \exp(j2\pi f_c t) \right\}$$

R: ground reflection coefficient (phase and amplitude change)

Two-path channel $h(t) = \delta(t) + 0.9\delta(t - \tau)$

- Channel parameters

$$\tau = 0.1T$$

- Delay spread

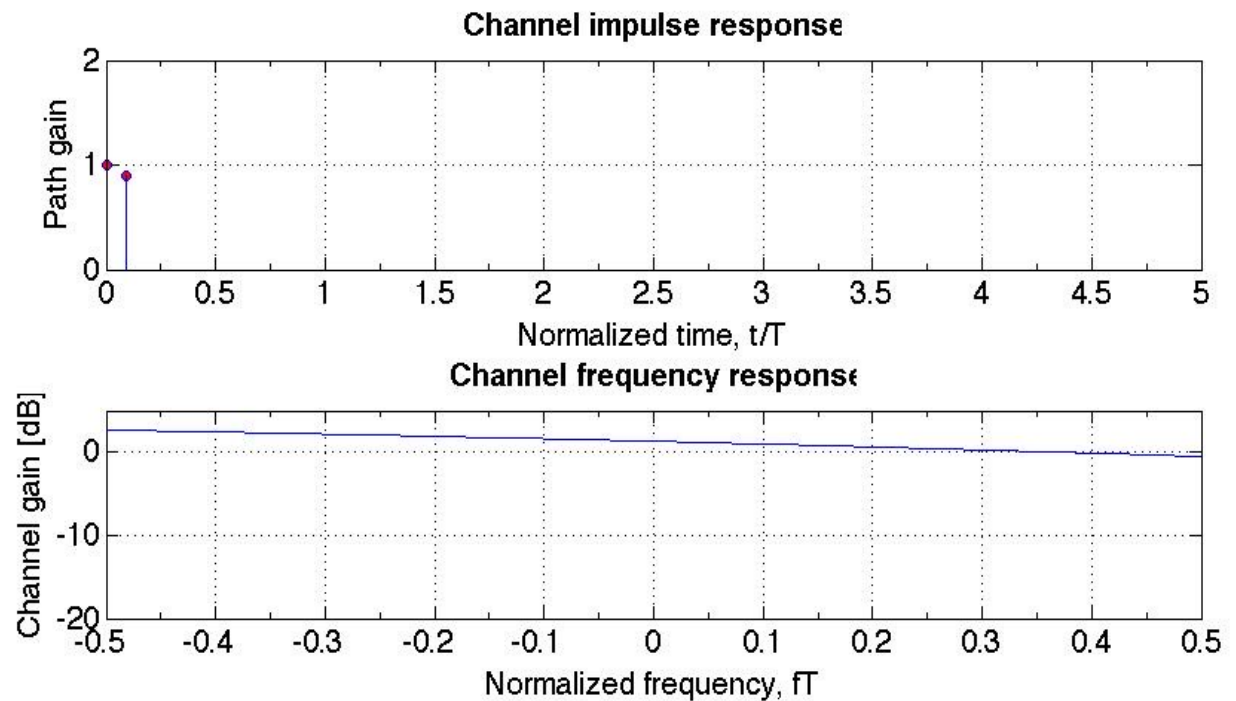
$$p_0 \approx 0.55$$

$$p_1 \approx 0.45$$

$$\bar{\tau} = 0.045T$$

$$\bar{\tau}^2 = 0.0045T^2$$

$$\sigma_\tau = \sqrt{\bar{\tau}^2 - (\bar{\tau})^2} = 0.05T$$



Two-path channel, $h(t) = \delta(t) + 0.9\delta(t - \tau)$

- Channel parameters

$$\tau = T$$

- Delay spread

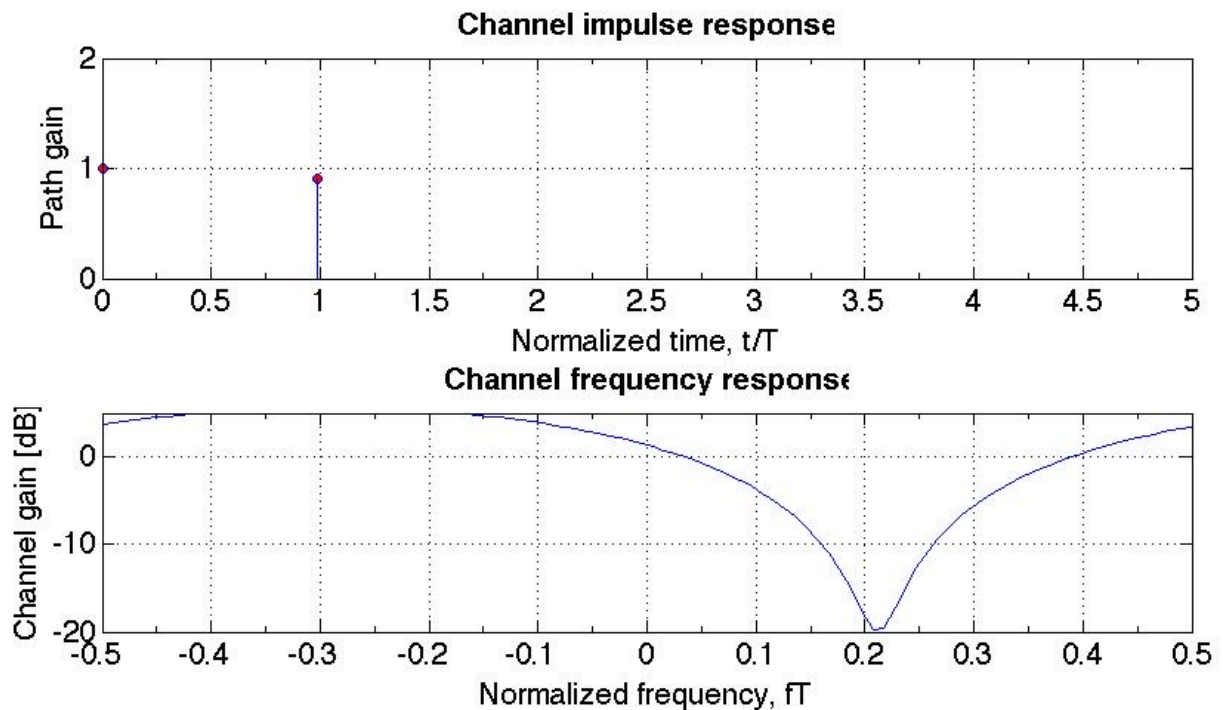
$$p_0 \approx 0.55$$

$$p_1 \approx 0.45$$

$$\bar{\tau} = 0.45T$$

$$\bar{\tau}^2 = 0.45T^2$$

$$\sigma_\tau = \sqrt{\bar{\tau}^2 - (\bar{\tau})^2} = 0.5T$$



Two-path channel, $h(t) = \delta(t) + 0.9\delta(t - \tau)$

- Channel parameters

$$\tau = 1.5T$$

- Delay spread

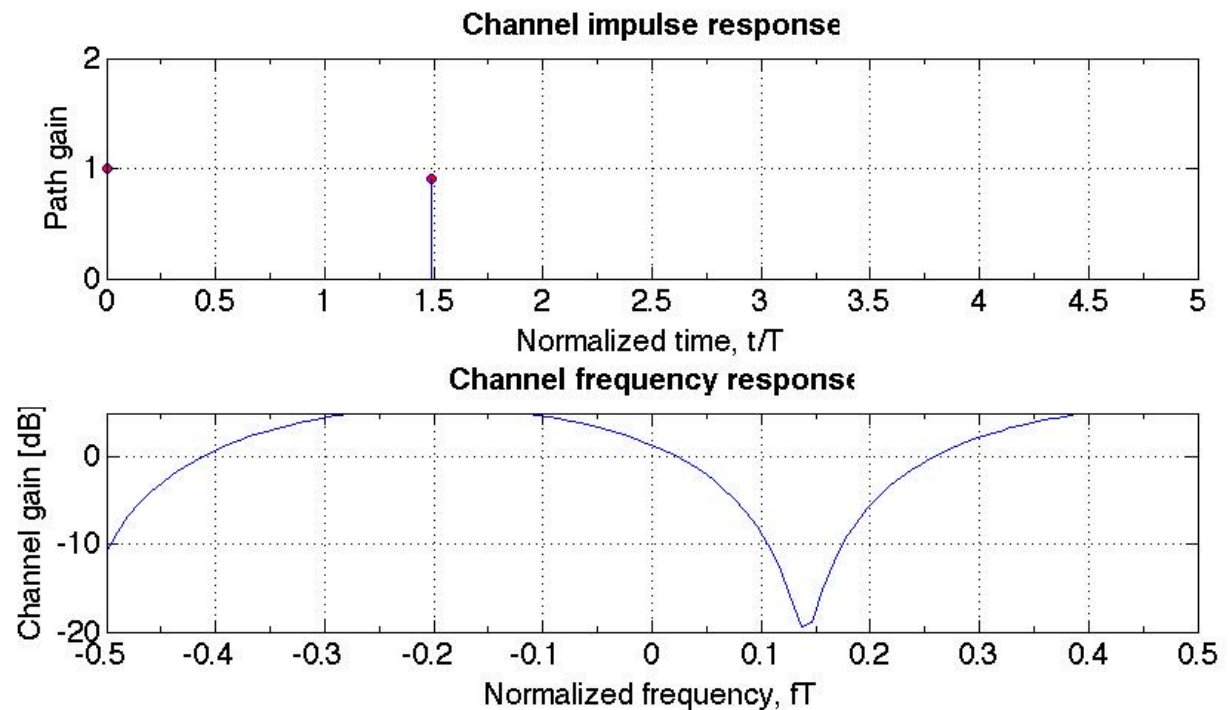
$$p_0 \approx 0.55$$

$$p_1 \approx 0.45$$

$$\bar{\tau} \approx 0.7T$$

$$\bar{\tau}^2 = T^2$$

$$\sigma_\tau = \sqrt{\bar{\tau}^2 - (\bar{\tau})^2} \approx 0.6T$$



Two-path channel, $h(t) = \delta(t) + 0.9\delta(t - \tau)$

- Channel parameters

$$\tau = 4T$$

- Delay spread

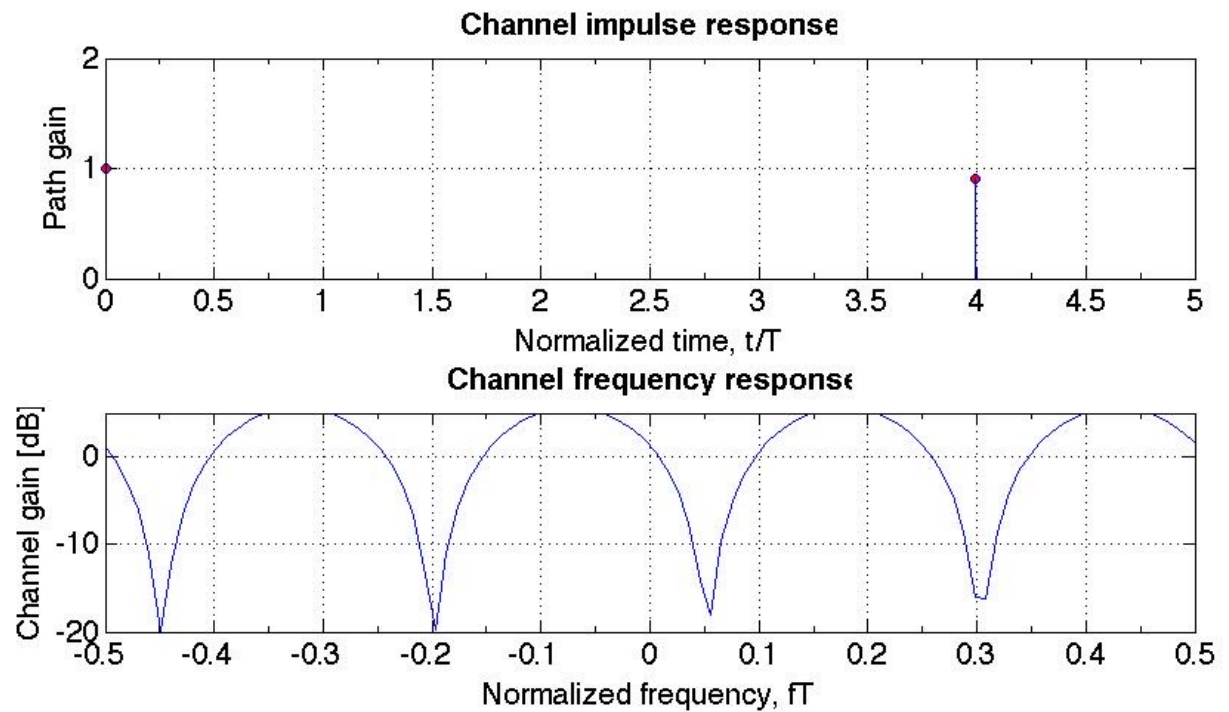
$$p_0 \approx 0.55$$

$$p_1 \approx 0.45$$

$$\bar{\tau} = 1.8T$$

$$\bar{\tau}^2 = 7.2T^2$$

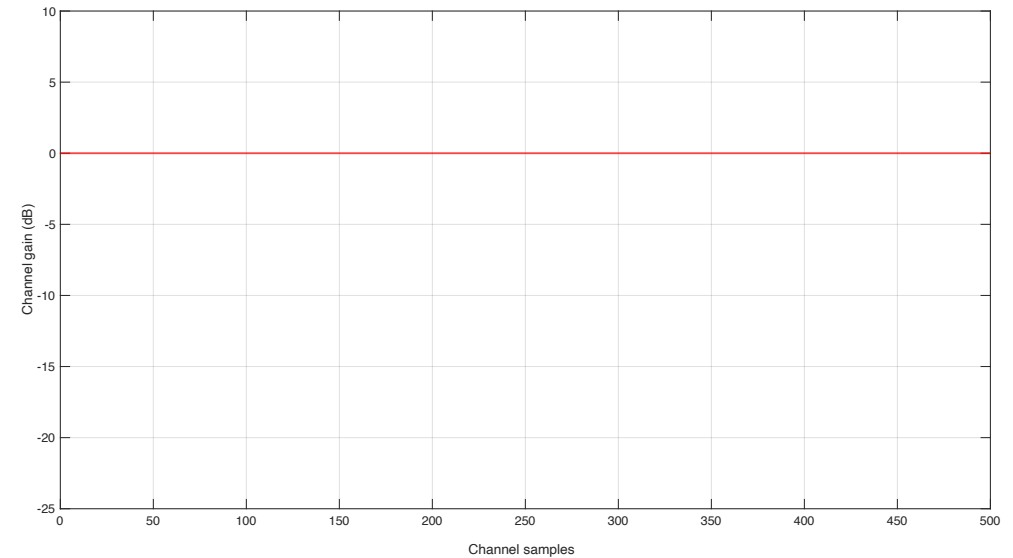
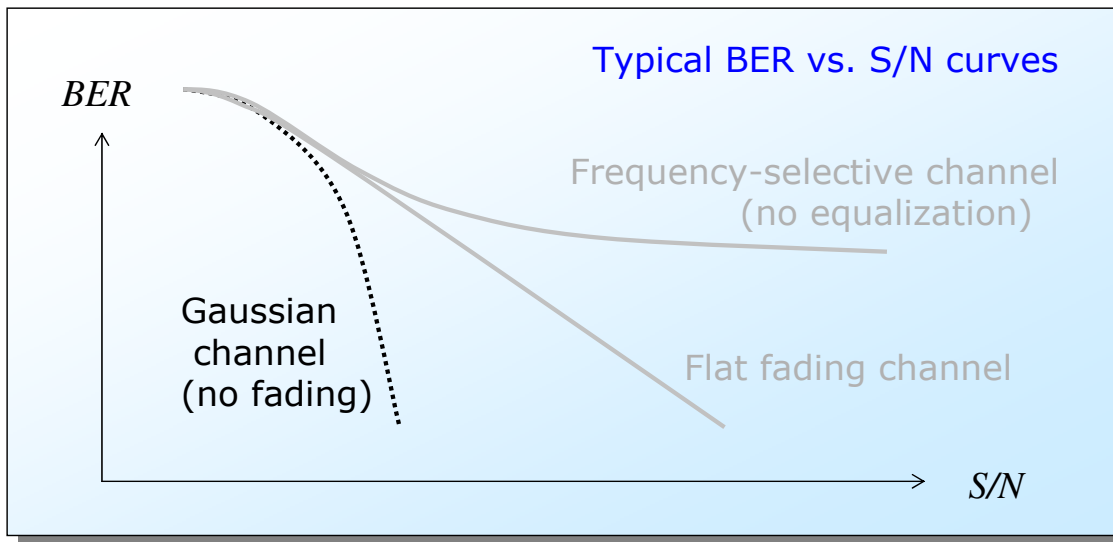
$$\sigma_\tau = T\sqrt{7.2 - (1.8)^2} \approx 2.3T$$



Flat fading channel: BER on AWGN

With an AWGN channel, the decision variable is

$$x(m) = c_m + n(m)$$



BER on flat Rayleigh fading channel

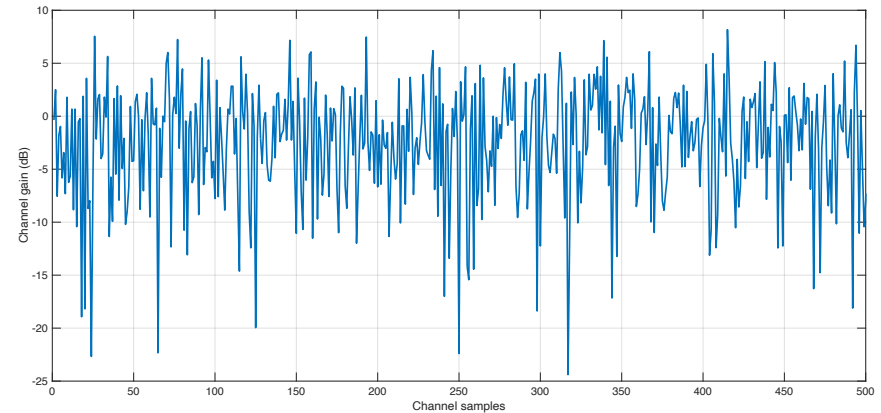
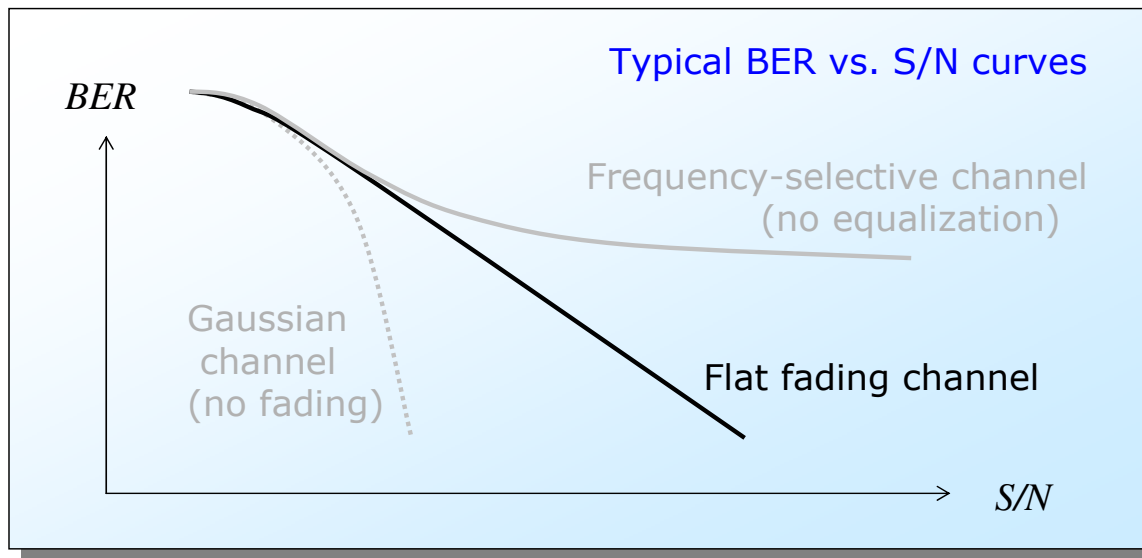
With a flat fading channel, the decision variable is

$$x(m) = \alpha c_m + n(m)$$

The channel α is Rayleigh distributed and it is $p(\alpha) = 2\alpha e^{-\alpha^2}$, so that $E\{\alpha^2\} = 1$.

The mean error probability is obtained by averaging the error probability over the channel

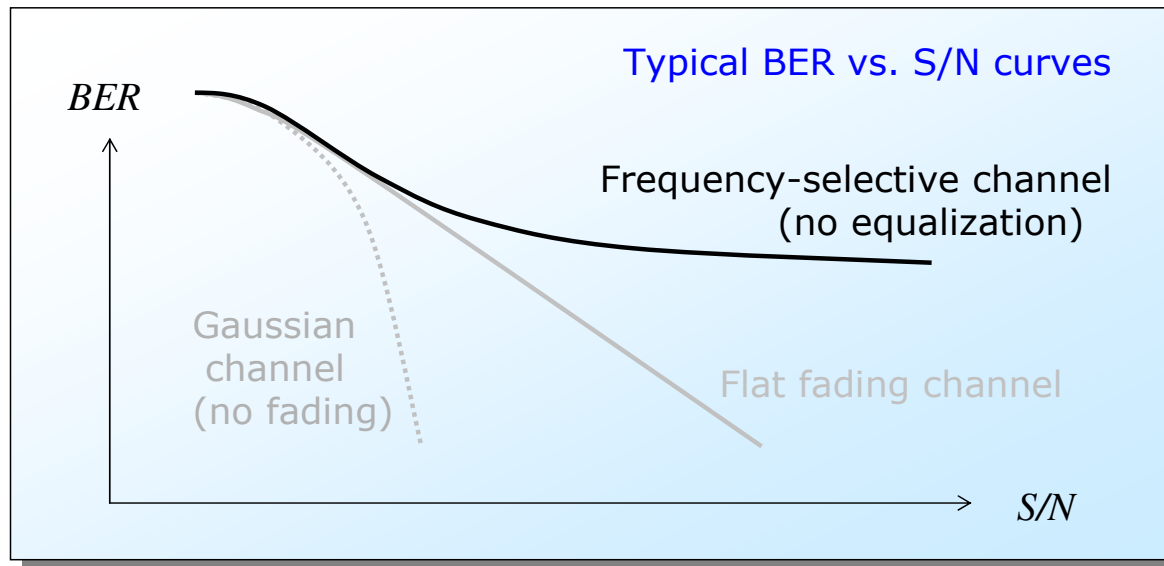
$$P_e = \int_0^{+\infty} P(e|\alpha)p(\alpha)d\alpha$$



BER on multipath Rayleigh fading channel

With a frequency selective channel, the decision variable is

$$x(m) = g(0)c_m + \underbrace{\sum_{k, k \neq 0} g(kT)c_{m-k}}_{\text{ISI}} + n(m)$$



If no countermeasures are taken, the error probability has an irreducible error-floor.

2-PAM BER on a flat fading channel

- On a flat fading channel, neglecting the symbol index the decision variable is

$$x = \alpha a + n$$

where α is a Rayleigh-distributed random variable and its pdf is

$$p(\alpha) = \frac{\alpha}{\sigma^2} e^{-\frac{\alpha^2}{2\sigma^2}},$$

- The msf of α is $E\{\alpha^2\} = 2\sigma^2$, so that when $\sigma^2 = \frac{1}{2} \Rightarrow E\{\alpha^2\} = 1$ and the pdf is

$$p(\alpha) = 2\alpha e^{-\alpha^2}$$

- For a given value of α , the conditional error probability is computed as

$$P(e|\alpha) = Q\left(\alpha \sqrt{\frac{2E_s}{N_0}}\right)$$

2-PAM BER on a flat fading channel

- The error probability is obtained by computing

$$P(e) = \int_0^{+\infty} p(\alpha)P(e|\alpha)d\alpha = \int_0^{+\infty} 2\alpha e^{-\alpha^2} Q\left(\alpha \sqrt{\frac{2E_s}{N_0}}\right) d\alpha$$

- Considered that $2\alpha e^{-\alpha^2} = \frac{d}{d\alpha} - e^{-\alpha^2}$, the integral can be solved by parts

$$\int_a^b f'(x)g(x)dx = f(x)g(x)\Big|_a^b - \int_a^b f(x)g'(x)dx.$$

- Since it is $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{+\infty} e^{-\frac{t^2}{2}} dt$, its derivative is

$$\frac{d}{dx} Q(x) = \frac{d}{dx} \frac{1}{\sqrt{2\pi}} \left[1 - \int_{-\infty}^x e^{-\frac{t^2}{2}} dt \right] = -\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

2-PAM BER on a flat fading channel

- Solving the integral by parts and replacing $\mu = \sqrt{\frac{2E_s}{N_0}}$ yields

$$\begin{aligned} P(e) &= -e^{-\alpha^2} Q(\alpha\mu) \Big|_0^{+\infty} - \int_0^{+\infty} (-e^{-\alpha^2}) \left(-\frac{\mu}{\sqrt{2\pi}} e^{-\frac{\alpha^2 \mu^2}{2}} \right) d\alpha \\ &= \frac{1}{2} - \frac{\mu}{\sqrt{2\pi}} \int_0^{+\infty} e^{-\frac{\alpha^2(2+\mu^2)}{2}} d\alpha, \end{aligned}$$

- Substituting $\beta = \alpha\sqrt{(2+\mu^2)}$ yields $d\alpha = \frac{1}{\sqrt{(2+\mu^2)}} d\beta$

$$P(e) = \frac{1}{2} - \frac{\mu}{\sqrt{(2+\mu^2)}} \frac{1}{\sqrt{2\pi}} \int_0^{+\infty} e^{-\frac{\beta^2}{2}} d\beta = \frac{1}{2} \left(1 - \frac{\mu}{\sqrt{(2+\mu^2)}} \right)$$

2-PAM BER on a flat fading channel

- Expressing $P(e)$ as a function of $\frac{E_s}{N_0}$, yields

$$P(e) = \frac{1}{2} \left(1 - \sqrt{\frac{\frac{E_s}{N_0}}{1 + \frac{E_s}{N_0}}} \right)$$

- When x is large it is $\sqrt{\frac{x}{1+x}} \approx 1 - \frac{1}{2x}$, and for large values of SNR the $P(e)$ can be approximated as

$$P(e) \approx \frac{1}{4 \frac{E_s}{N_0}}$$

Time-varying channel

- If the mobile receiver is in movement, the gains and the phase of the various paths of the channel vary in time

$$h(t, \tau) = A_{LS} \sum_{\ell=0}^{N_c-1} \alpha_{\ell}(t) e^{j\phi_{\ell}(t)} \delta(\tau - \tau_{\ell})$$

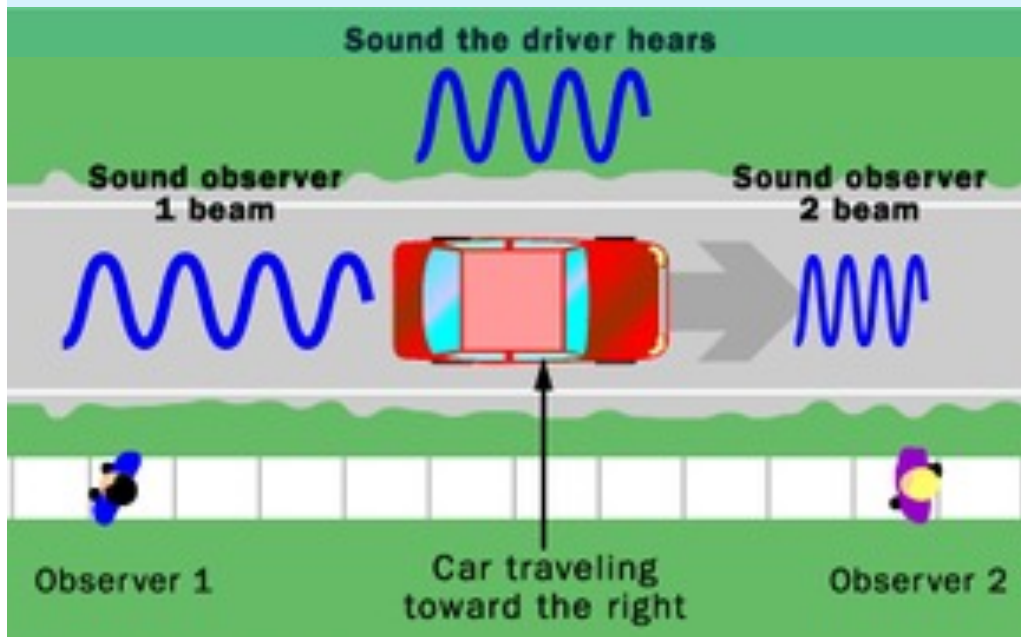
- The received signal is

$$y(t) = s(t) \otimes h(t, \tau) = A_{LS} \sum_{\ell=0}^{N_c-1} \alpha_{\ell}(t) e^{j\phi_{\ell}(t)} s(t - \tau_{\ell})$$

- The channel gains and phases change much faster than the large-scale fading A_{LS} and the delays τ_{ℓ} .

The Doppler effect: toy example

- Speed of sound: $343\text{m/s} \approx 1200\text{km/h}$.
- Distance between Obs1 and Obs2 is $d = 2\text{km}$.
- The ambulance travels at 120 km/h and takes 60s to go from Obs1 to Obs2.



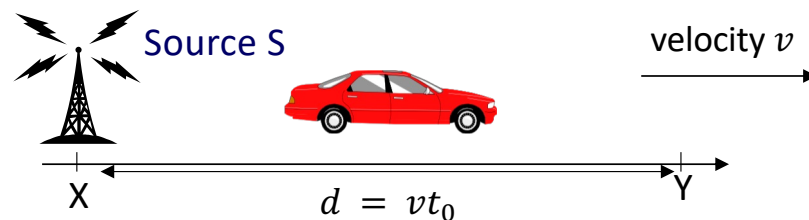
- At t_0 , the ambulance is at the position of Obs1 and starts the syren. The sound takes 6s to travel the distance d .
- At $t_0 + 6$, Obs2 hears the syren.
- At $t_0 + 60$, the ambulance is at the position of Obs2 and stops the syren.
- At $t_0 + 66$, Obs1 stops hearing the syren.

The duration of the syren is 54s for Obs2, 60s for the ambulance's driver and 66s for Obs 1.

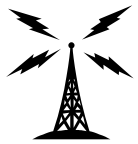
Doppler shift

- Consider a mobile user moving at constant velocity v between points X and Y.
- At $t = 0$ the user is in X and the source S, also located in X, starts transmitting a sinusoidal signal $s(t) = \sin(2\pi f_c t)$.
- At time $t = t_0$, the mobile user is in Y having travelled a distance $d = vt_0$, which the radio signal takes the time $\Delta\tau = vt_0/c$ to travel.
- At time $t = t_0$, the received signal at point X is $y_X(t_0) = \sin(2\pi f_c t_0)$, while the received signal at point Y is

$$\begin{aligned} y_Y(t_0) &= \sin(2\pi f_c(t_0 - \Delta\tau)) \\ &= \sin(2\pi f_c t_0 - f_c vt_0/c) = \sin(2\pi(f_c - f_c v/c)t_0) \end{aligned}$$



Doppler shift



distance $d = vt_0$

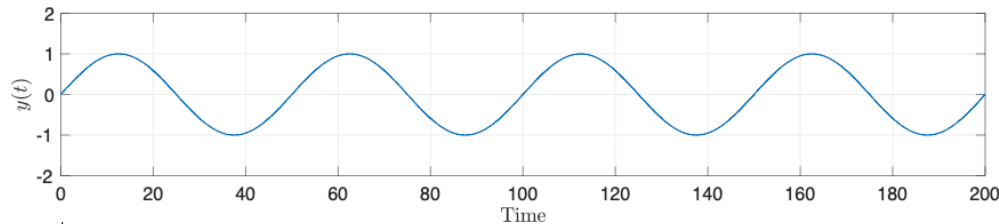


velocity v

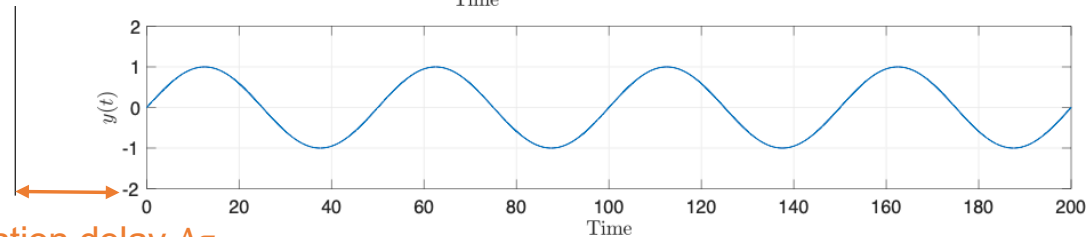
Mobile moving away from base $\Rightarrow v > 0$, Doppler shift < 0

Mobile moving towards base $\Rightarrow v < 0$, Doppler shift > 0

Received signal $y(t)$
at point X



Received signal $y(t)$
at point Y



Propagation delay $\Delta\tau$

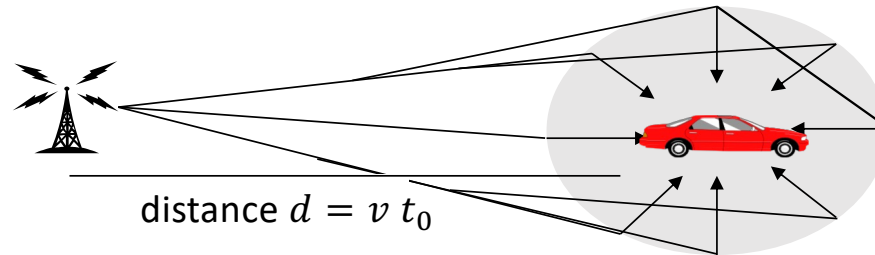
$$y_Y(t_0) = \sin 2\pi f_c (t_0 - \Delta\tau) = \sin 2\pi \left(f_c - \frac{f_c v}{c} \right) t_0$$

received frequency

$$\text{Doppler shift } f_D = -\frac{f_c v}{c} = -\frac{v}{\lambda}$$

Multipath channel: Doppler Spectrum

- In fading channels many signal replicas arrive at the receiver with different angles. The effect is a *Doppler spread* rather than a single shift.



- The received signal is the sum of all scattered waves and is not anymore deterministic but is modeled as a *stochastic process* described by its *autocorrelation* and *power spectral density*.
- Assuming that the received energy is the same from all directions (uniform scattering), the time-varying channel is modelled by the *Jakes' Doppler spectrum*.

Time varying channels as a stochastic process

- The Doppler spectrum

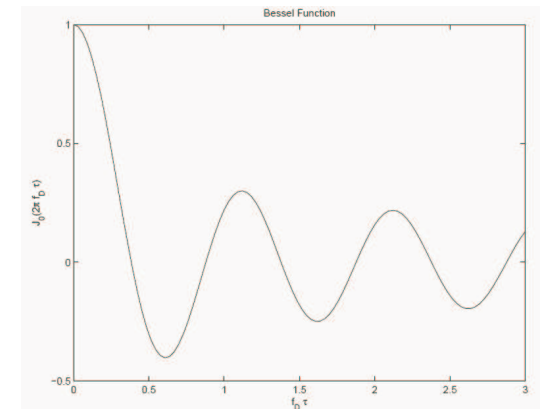
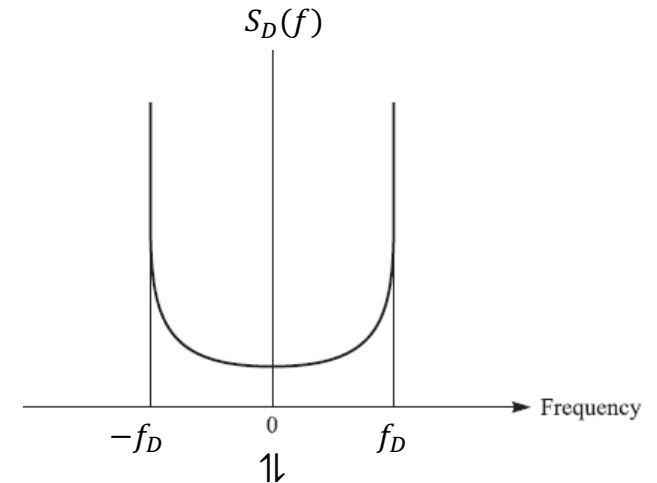
$$S_D(f) = \frac{1}{\pi f_D} \left(\sqrt{1 - \left(\frac{f}{f_D} \right)^2} \right)^{-1}$$

is the power spectral density of the time varying channel.

- In the time domain, the autocorrelation function of the channel is

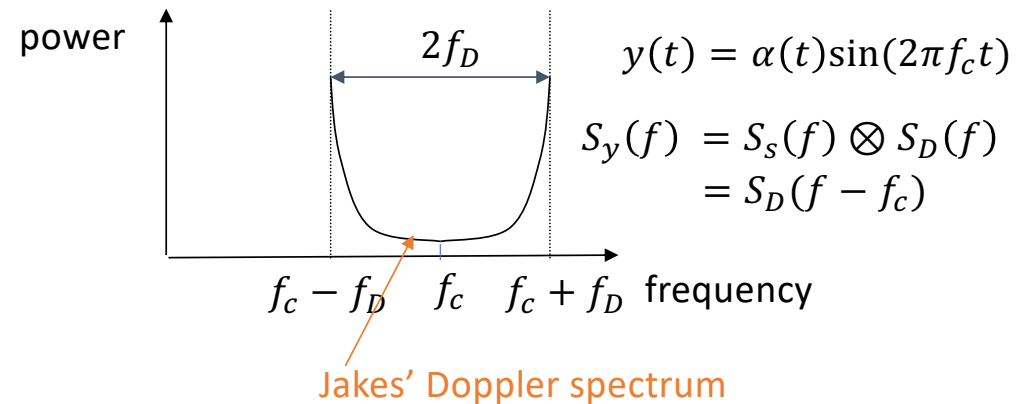
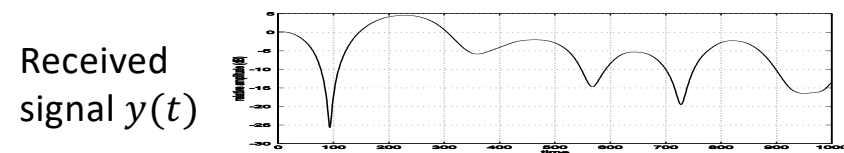
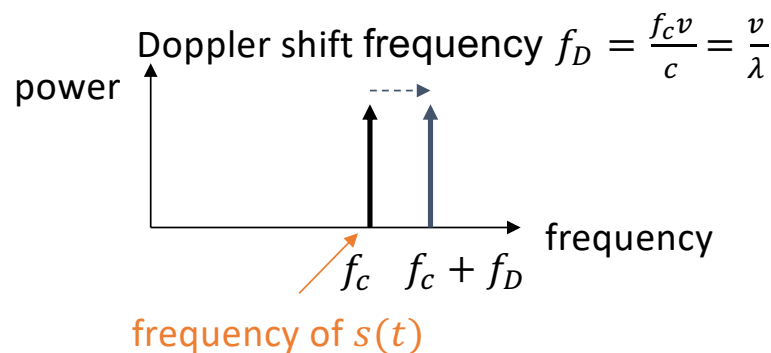
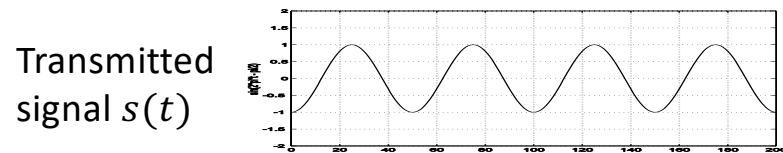
$$\rho(t) = J_0(2\pi f_D t) \Leftrightarrow S_D(f)$$

where $J_0(x)$ is the 0th-order Bessel function of the first kind.



Jakes' Doppler spectrum for a sinusoidal tone

- The spectral broadening caused by the receiver movement is called *Doppler spread*.



Multipath channel: Doppler Spectrum

- The effect of mobility in a multipath environment is that the channel is time-varying.
- In particular, setting $A_{LS} = 1$

- If the channel is *frequency flat*, the received signal is

$$y(t) = \alpha(t)e^{j\phi(t)}s(t - \tau_\ell)$$

- If the channel is *frequency selective*, the received signal is

$$y(t) = \sum_{\ell=0}^{N_c-1} \alpha_\ell(t)e^{j\phi_\ell(t)}s(t - \tau_\ell)$$

- In any case, the the PSD of each channel path is modelled by the Jakes' Doppler spectrum.

Jakes' Doppler spectrum – flat fading channel

- The effect of mobility is a *broadening* of the signal spectrum and the PSD of the received signal

$$y(t) = a(t)s(t)$$

is obtained as the convolution of the signal power spectral density $S_s(f)$ and the Jakes Doppler spectrum $S_D(f)$, i.e.

$$S_y(f) = S_s(f) \otimes S_D(f)$$

- In particular, if the transmitted signal is a sinusoid, i.e. $s(t) = \sin(2\pi f_c t)$, the PSD of the received signal $y(t)$ is

$$S_y(f) = S_s(f) \otimes S_D(f) = \frac{1}{\pi f_D} \left(\sqrt{1 - \left(\frac{f - f_c}{f_D} \right)^2} \right)^{-1}$$

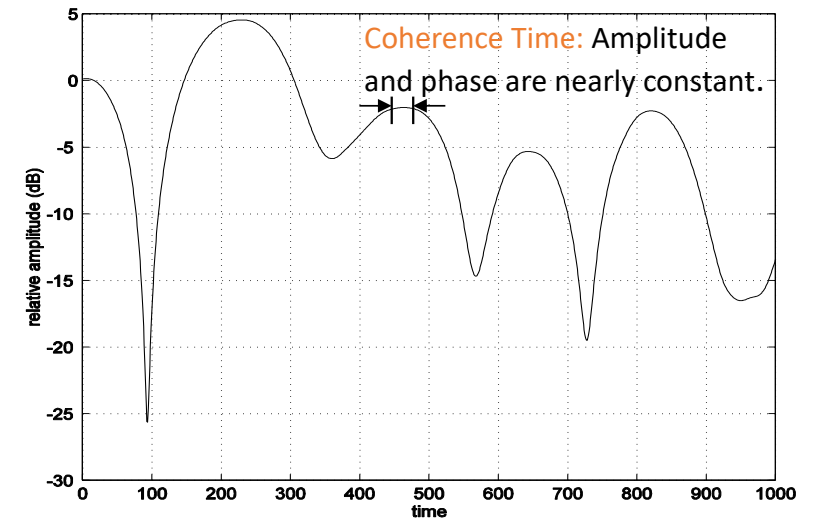
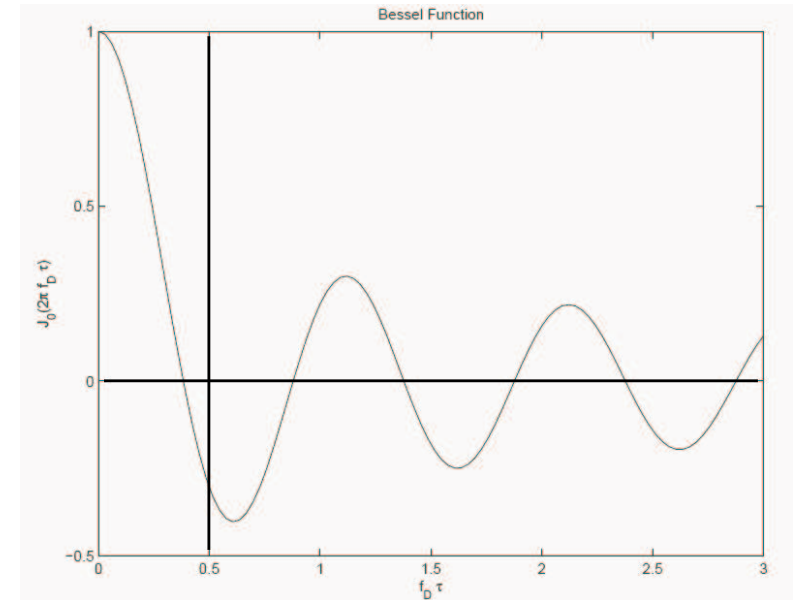
Channel's coherence time

- $J_0(2\pi x) \approx 0$ for $x = \frac{1}{2} \Rightarrow$ The channel can be assumed uncorrelated for $f_D T_c = \frac{1}{2}$.
- The *channel's coherence time* T_c is defined as the time interval over which the channel can be approximated as constant.

$$T_c = \frac{1}{2f_D}$$

- In terms of distance, the *coherence distance* is

$$d_c = vT_c = \frac{1}{2}v \frac{c}{f_c v} = \frac{\lambda}{2}$$



Doppler spectrum: slow and fast fading

- Doppler spread is a measure of the *spectral broadening* caused by motion.
 - If the baseband signal bandwidth $B_s \gg f_D$ then the effect of Doppler spread is negligible at the receiver and the channel is *slow fading*.
 - If $B_s < f_D$ then the channel is *fast fading* and the Doppler spread severely distorts the received signal, which often results in an irreducible BER and synchronization problems.
- Similar considerations can be made in terms of symbol duration
 - A channel is *slow fading* if $T_c > T$.
 - A channel is said to be *fast fading* if $T_c < T$.

Fading channel example

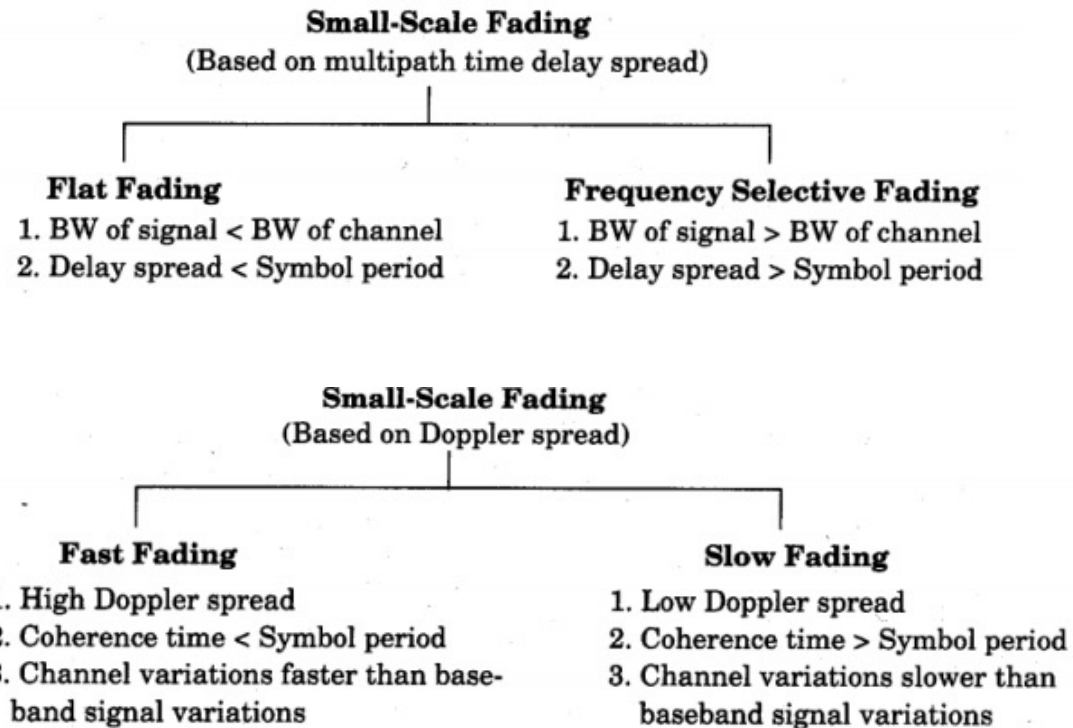
- Consider a transmission at $f_c = 2.1$ GHz in a suburban area (delay spread $\sigma_\tau = 2 \mu s$) to a user moving at a speed 90 km/h $\Rightarrow v = 25$ m/s. The signal bandwidth is $B_s = 2$ MHz $\Rightarrow$ the symbol time can be approximated as $T \sim \frac{1}{B_s} = 500$ ns.

- The Doppler shift is

$$f_D = \frac{f_c v}{c} = \frac{2.1 \cdot 10^9 \cdot 25}{3 \cdot 10^8} = 175 \text{ Hz} \Rightarrow T_c = \frac{1}{2f_d} \sim 3 \text{ ms.}$$

- The channel coherence bandwidth is $B_c = \frac{1}{5\sigma_\tau} = 100$ kHz.
- The channel is *slow* ($B_s \gg f_d$ or $T \ll T_c$) and *frequency-selective* ($B_s > B_c$ or $T < \sigma_\tau$).

Small-scale fading recap



Small-scale fading recap

